FathomSplat: Physics-Guided Feed-Forward 3D Gaussian Splatting for Underwater Reconstruction

Anonymous submission

Abstract

Underwater 3D Gaussian splatting achieves photorealistic reconstruction by explicitly modeling attenuation and backscatter, but relies on structure-from-motion initialization and costly per-scene optimization, limiting its use in latency-sensitive robotics. Feed-forward reconstruction offers a promising alternative by removing test-time optimization, yet current models are built on clear-air image formation and cannot account for wavelength-dependent attenuation or backscatter, which corrupt appearance cues, entangle scene geometry with medium effects, and cause range-dependent attenuation to be read as depth. Explicit medium modeling and amortized feed-forward inference have thus remained largely incompatible. We present *FathomSplat*, a physics-guided feed-forward 3D Gaussian splatting framework for fast underwater scene reconstruction, which jointly predicts camera parameters, a medium-free 3D Gaussian representation, and per-view medium parameters from sparse, unposed underwater images. A differentiable rewatering operator applies a physical underwater image-formation model to medium-free renderings, maps them into the observation domain, and regularizes the inherently ambiguous decomposition of scene structure and medium effects. Paired synthetic supervision and real-data adaptation enable direct medium-free rendering without structure-from-motion or scene-specific optimization. On held-out FLSea with 16 input views, FathomSplat achieves 21.92 dB PSNR, 0.616 SSIM, and 0.297 LPIPS in 0.20 s.

1 Introduction

Reliable three-dimensional perception is fundamental to underwater robotic inspection, environmental survey, navigation, and manipulation. These applications require scene representations that preserve meaningful geometry, support high-quality novel-view rendering, and can be constructed with low latency. Among recent representations, 3D Gaussian Splatting (3DGS) is particularly attractive because its explicit scene structure enables efficient, photorealistic rendering (Kerbl et al. 2023). Underwater neural rendering methods further incorporate physical image formation into NeRF or 3DGS and jointly estimate scene and medium properties for each capture (Levy et al. 2023; Li et al. 2025; Yang, Leonard, and Girdhar 2025; Mualem et al. 2026; Wang et al. 2025a). Their results show that explicit medium modeling can recover high-quality underwater scenes while accounting for attenuation and backscatter.

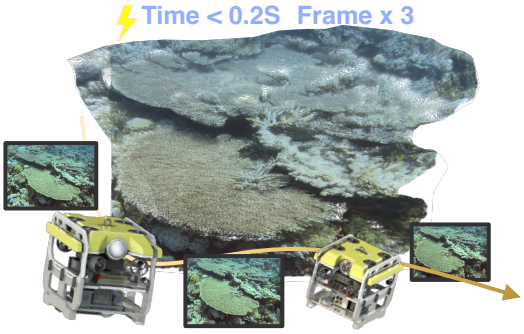


Figure 1: FathomSplat is the first feed-forward underwater 3DGS model and runs in under 0.2 s with three views.

Despite efficient rendering, existing underwater 3DGS methods retain a costly reconstruction process. They optimize a separate representation for every capture and typically depend on calibrated cameras or structure-from-motion initialization such as COLMAP (Schönberger and Frahm 2016). This repeated computation is poorly suited to sparse-view, online operation. Feed-forward reconstruction offers a promising alternative by replacing scene-specific optimization with amortized prediction. Geometry foundation models recover dense scene structure and, in recent systems, camera parameters from unposed image collections (Wang et al. 2024; Leroy, Cabon, and Revaud 2024; Wang et al. 2025b, 2026). A complementary line of work directly predicts renderable 3D Gaussian scenes from sparse images. PixelSplat (Charatan et al. 2024) and MVSplat (Chen et al. 2024) assume known camera parameters, while NoPoSplat (Ye et al. 2025) and AnySplat (Jiang et al. 2025) operate on unposed image collections. These advances make low-latency reconstruction increasingly practical, but their image-formation assumptions remain designed for clear-air scenes.

Existing feed-forward models treat observed radiance primarily as a surface-attached attribute whose variation is explained by reflectance, illumination, and viewpoint. Under-

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water image formation violates this assumption. Wavelength-dependent attenuation suppresses the direct signal as camera-to-surface range increases, while backscatter contributes radiance along the line of sight that is generated by the medium rather than by a physical surface (Akkaynak and Treibitz 2018). Consequently, surface appearance varies systematically with range and water properties, and part of the measured image cannot be assigned to scene geometry. Under conventional photometric objectives, a feed-forward model may absorb color casts and haze into Gaussian appearance and entangle range-dependent attenuation with predicted geometry. Existing methods therefore either perform feed-forward reconstruction under clear-air image formation or model underwater media through scene-specific optimization. To our knowledge, no prior method predicts both a renderable medium-free 3D representation and the medium parameters explaining sparse underwater observations without test-time optimization.

We address this gap with *FathomSplat*, a physics-guided feed-forward framework for reconstructing medium-free 3D Gaussian scenes from sparse underwater views (Figure 1). FathomSplat jointly predicts camera parameters, scene geometry and appearance, and per-view medium parameters. The formulation separates scene and medium: multi-view and depth cues anchor scene structure, while a differentiable *rewatering* operator explains underwater observations from predicted clear radiance and medium parameters. Paired synthetic supervision constrains this otherwise ambiguous decomposition, allowing the operator to be omitted for direct medium-free rendering at inference. We then adapt the model to real underwater imagery using rewatering consistency, relative-geometry supervision, and synthetic replay.

Our main contributions are:

- To our knowledge, FathomSplat is the first medium-aware feed-forward framework for underwater 3D Gaussian reconstruction. It reconstructs sparse underwater views without per-scene optimization or SfM at inference and is designed to generalize across scenes and water conditions.
- We introduce an explicit medium–scene factorization in which a medium-free Gaussian representation and predicted water parameters are coupled by a differentiable image-formation operator. By assigning attenuation and backscatter to a separate observation model, the factorization reduces the incentive to absorb range-dependent medium effects into Gaussian appearance and geometry while directly supporting medium-free rendering.
- We develop a synthetic-to-real training strategy that constrains the otherwise underdetermined decomposition through direct supervision of clear appearance, rewatered observations, medium parameters, and geometry, followed by real-domain adaptation with relative-geometry supervision and synthetic replay.

2 Related Work

Feed-forward 3D reconstruction and view synthesis. Feed-forward reconstruction amortizes geometric inference across scenes rather than optimizing each scene independently. Geometry-first systems such as DUSt3R (Wang et al. 2024), MASt3R (Leroy, Cabon, and Revaud 2024), VGGT (Wang et al. 2025b), and VGGT-Ω (Wang et al. 2026) infer dense structure and camera parameters from image collections. A complementary line directly predicts renderable 3D Gaussian scenes (Kerbl et al. 2023). PixelSplat (Charatan et al. 2024) and MVSplat (Chen et al. 2024) assume posed inputs, Splatt3R (Smart et al. 2024) handles uncalibrated image pairs, and NoPoSplat (Ye et al. 2025), FLARE (Zhang et al. 2025), and AnySplat (Jiang et al. 2025) operate on sparse unposed views. These models are designed primarily for clear-air image formation and provide no explicit mechanism for range-dependent attenuation or radiance generated by the medium.

Underwater neural rendering and reconstruction. Underwater neural rendering incorporates scattering-aware image formation into 3D reconstruction. SeaThru-NeRF (Levy et al. 2023) separates object radiance from medium contributions, while U2NeRF (Gupta et al. 2024) combines restoration and novel-view synthesis from posed observations. Underwater 3DGS methods model the medium through volumetric fields or explicit scene and medium factorization. WaterSplatting (Li et al. 2025) couples Gaussians to a volumetric medium field; SeaSplat (Yang, Leonard, and Girdhar 2025), Gaussian Splashing (Mualem et al. 2026), UW-GS (Wang et al. 2025a), and 3D-UIR (Yuan et al. 2026) separate scene appearance from underwater degradation. AquaSplatting (Hu et al. 2026) and WaterClear-GS (Zhang et al. 2026) further parameterize geometry and optical properties, while RestorGS (Qiao et al. 2025) and OceanSplat (Kweon and Park 2026) focus on appearance decoupling and geometry regularization. Despite their different formulations, these methods optimize a new scene for each capture and generally use known or separately estimated cameras. FathomSplat instead amortizes reconstruction across scenes and directly predicts cameras, a renderable medium-free Gaussian scene, and the medium parameters explaining sparse underwater observations, without test-time optimization.

Underwater image formation and restoration. Underwater image formation follows the scattering models of Jaffe and McGlamery (McGlamery 1980; Jaffe 1990) and later refinements. The revised model (Akkaynak and Treibitz 2018) distinguishes wavelength-dependent direct attenuation from backscatter attenuation. Sea-thru (Akkaynak and Treibitz 2019) recovers scene radiance when range is available, while DeepSeeColor (Jamieson, How, and Girdhar 2023) predicts attenuation and backscatter from an RGB image and corresponding depth. Both rely on independently supplied geometry and target image restoration rather than consistent multi-view reconstruction and camera estimation. These methods motivate our observation model but do not couple recovered appearance to joint 3D scene estimation.

3 Method

FathomSplat amortizes underwater inverse rendering into a single network that jointly estimates cameras, a medium-free 3D Gaussian scene, and the intervening water medium.

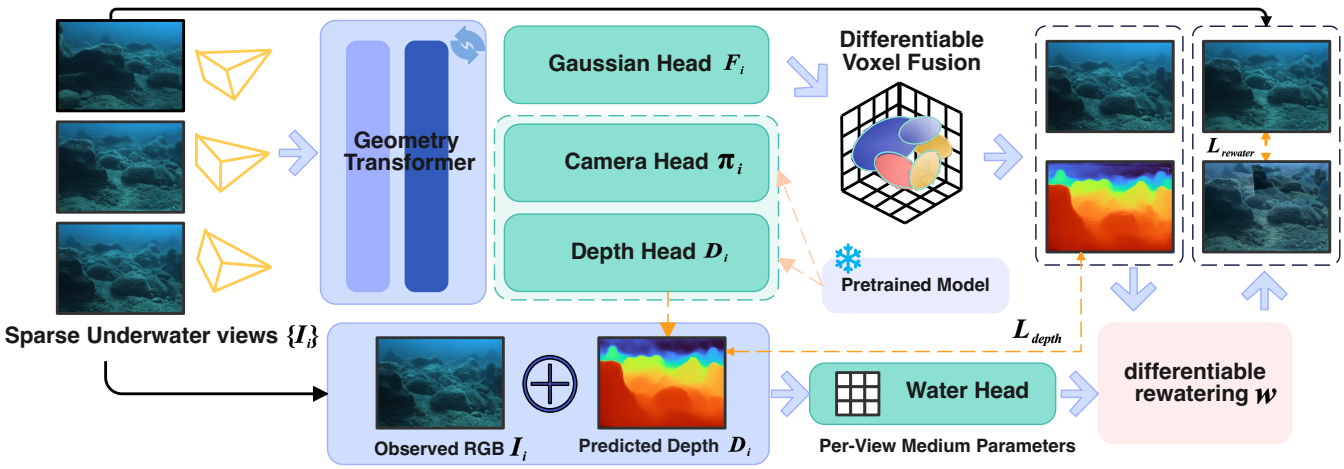


Figure 2: Overview of FathomSplat. A single forward pass predicts cameras, a medium-free 3D Gaussian scene, and per-view water parameters. During training, differentiable rewatering maps the medium-free render back to the observation domain. At inference, the same Gaussian representation is rendered directly without the medium operator.

Figure 2 summarizes the framework. We first formulate the feed-forward reconstruction problem and scene representation (Secs. 3.1–3.2), then introduce the physics-guided medium–scene factorization (Sec. 3.3) and its paired supervision (Sec. 3.4). Finally, we describe real-domain adaptation (Sec. 3.5) and the progressive training protocol (Sec. 3.6).

3.1 Overview and Problem Formulation

Let $\mathcal{I} = \{I_i\}_{i=1}^N$ denote N underwater images with unknown intrinsics and extrinsics. We seek a renderable scene representation whose appearance is independent of the water medium, together with the medium parameters required to explain the observations. The scene is represented by 3D Gaussians $\mathcal{G} = \{(\mu_k, \alpha_k, s_k, q_k, c_k)\}_{k=1}^K$, where μ_k, α_k, s_k, q_k, and c_k denote center, opacity, scale, rotation, and view-dependent appearance, respectively. For each input view, the medium is parameterized by $\theta_i = (\beta_i^D, \beta_i^B, B_i^\infty)$. A network f_ψ jointly predicts the cameras $\Pi = \{\pi_i\}$, scene Gaussians, and medium parameters:

$$(\hat{\Pi}, \mathcal{G}, \hat{\Theta}) = f_\psi(\mathcal{I}), \qquad \hat{\Theta} = \{\hat{\theta}_i\}_{i=1}^N. \tag{1}$$

Differentiable Gaussian rasterization (Kerbl et al. 2023) at camera π yields a medium-free color image and depth map, $\mathcal{R}(\mathcal{G}, \pi) = (\hat{J}(\pi), \hat{z}(\pi))$. The predicted camera parameters eliminate the need for SfM at inference, and all quantities in Eq. 1 are obtained without scene-specific optimization.

3.2 Feed-Forward Reconstruction Backbone

Geometry backbone. We use VGGT-Ω (Wang et al. 2026) to encode the input collection jointly. A frozen DINOv3 patch-16 tokenizer extracts per-view tokens, after which the cross-view aggregator predicts dense features F_i, depth D_i, and camera parameters π_i. Joint processing allows geometry to be inferred from cross-view correspondences and learned geometric priors; attenuation is not used as an explicit range estimator.

Gaussian Head. Following AnySplat (Jiang et al. 2025), a DPT-style Gaussian Head maps F_i to per-pixel opacity, covariance, and spherical-harmonic appearance parameters. The predicted depth is unprojected with π_i to obtain Gaussian centers in a common coordinate frame. Confidence-weighted differentiable voxel fusion merges redundant cross-view primitives into the scene representation $\mathcal{G}$. Unlike a direct underwater adaptation of AnySplat, the appearance parameters of $\mathcal{G}$ are supervised in the medium-free domain; range-dependent water effects are assigned to the separate medium branch.

3.3 Physics-Guided Decomposition

Underwater image formation. We adopt the revised underwater image-formation model of Akkaynak and Treibitz (2018). For color channel $c \in \{R, G, B\}$, medium-free radiance J_c, and camera-to-surface range z, define the rewatering operator

$$\mathcal{W}_c(J, z; \theta) = J_c e^{-\beta_c^D z} + B_c^\infty \left(1 - e^{-\beta_c^B z}\right). \tag{2}$$

Here, β_c^D controls wavelength-dependent attenuation of the direct signal, whereas β_c^B and B_c^∞ govern the growth and asymptotic radiance of backscatter. Maintaining separate coefficients for the two components is important because their effective attenuation differs in real water.

Water-Head. A lightweight Water-Head estimates $\hat{\theta}_i$ independently for each input view. It receives the observed RGB image concatenated with $\log(1 + D_i)$ and uses a convolutional encoder with global pooling to predict nine values: the three RGB channels of β^D, β^B, and B^∞. Bounded activations enforce non-negative attenuation coefficients and $B^\infty \in [0, 1]$. This global-per-image parameterization corresponds to a homogeneous medium within each view while permitting moderate variation across observations. For a held-out target view whose image is unavailable, we form a scene-level estimate $\bar{\theta}$ by taking the component-wise median

of the parameters predicted from the context views. Thus target-view rendering uses context evidence only; it does not estimate medium parameters from the target RGB. This aggregation assumes that the context and target belong to the same approximately homogeneous water body. A spatial variant can replace the global pooling when heterogeneous media are considered.

Differentiable rewatering. For a supervised input camera π_i, the predicted underwater image is

$$\hat{I}_i = \mathcal{W}\Big(\hat{J}(\pi_i), \hat{z}(\pi_i); \hat{\theta}_i\Big). \tag{3}$$

For an unseen target camera π_t, observed-domain NVS instead uses $\mathcal{W}(\hat{J}(\pi_t), \hat{z}(\pi_t); \bar{\theta})$, where $\bar{\theta}$ is computed exclusively from context views. Equations 2–3 establish an analysis-by-synthesis path in the observation domain. Explicitly accounting for attenuation and backscatter reduces the incentive to encode range-dependent color in Gaussian appearance or to represent backscatter with unsupported geometry. The factorization alone remains underdetermined, however; the paired supervision introduced next is therefore essential. For medium-free inference, $\mathcal{W}$ is omitted and $\mathcal{G}$ is rendered directly.

3.4 Paired Physics Supervision

The decomposition cannot be uniquely recovered from an underwater observation alone. We therefore construct paired synthetic training data from clear RGB-D sequences. For each image, medium parameters θ_{gt} are sampled from physically plausible, channel-dependent ranges and used to generate $I_{gt} = \mathcal{W}(J_{gt}, z_{gt}; \theta_{gt})$. The model receives I_{gt} and is supervised in both the latent medium-free domain and the observation domain:

$$\mathcal{L}_{clear} = \|\hat{J} - J_{gt}\|_2^2, \tag{4}$$

$$\mathcal{L}_{rewater} = \|\hat{I} - I_{gt}\|_2^2 + \lambda_p \,\mathrm{LPIPS}(\hat{I}, I_{gt}), \tag{5}$$

$$\mathcal{L}_{medium} = \|\hat{\theta} - \theta_{gt}\|_1. \tag{6}$$

$\mathcal{L}_{clear}$ anchors Gaussian appearance to the clear image, $\mathcal{L}_{rewater}$ enforces consistency with the measured domain, and $\mathcal{L}_{medium}$ directly supervises the physical parameters known from synthesis. Geometry is constrained by a scale-invariant ℓ_1-plus-gradient loss $\mathcal{L}_{GT}$ between backbone depth D and z_{gt}, together with $\mathcal{L}_{depth}$ between the rasterized depth $\hat{z}$ and D. A frozen VGGT-Ω teacher provides pose regularization $\mathcal{L}_{pose}$; teacher depth and normal predictions are not used in the principal configuration. The synthetic objective is

$$\begin{aligned}\mathcal{L} = \lambda_c\mathcal{L}_{clear} + \lambda_r\mathcal{L}_{rewater} + \lambda_m\mathcal{L}_{medium} \\ + \lambda_g\mathcal{L}_{GT} + \lambda_d\mathcal{L}_{depth} + \lambda_o\mathcal{L}_{pose}.\end{aligned} \tag{7}$$

Because $\mathcal{L}_{GT}$ median-aligns predicted and ground-truth depth, it constrains relative shape rather than metric scale. Sampling θ_{gt} independently across training views exposes the predictor to diverse water conditions and discourages specialization to a single medium.

3.5 Real-Domain Adaptation

The model in Eq. 2 does not capture all effects present in operational imagery, including forward scattering, non-uniform artificial illumination, and suspended particles. We therefore perform domain-level adaptation on real underwater training sequences. In the absence of clear targets and medium labels, the RGB signal is supervised through $\mathcal{L}_{rewater}$, while synthetic batches are replayed to retain the decomposition learned from paired data.

Where reliable depth supervision is available, we retain the scale-invariant geometry objective used during paired training and apply it only on valid pixels. This supervision constrains relative scene shape without introducing an absolute-scale claim. The resulting domain-adapted checkpoint is frozen when evaluated on held-out scenes, preserving feed-forward inference without test-scene optimization.

3.6 Progressive Training

Training proceeds in three stages. First, we establish the feed-forward Gaussian predictor on diverse clear-air datasets, initially optimizing the Gaussian Head and subsequently updating the geometry backbone at a reduced learning rate. Second, starting from the same clear-air checkpoint, we initialize the Water-Head and optimize Eq. 7 on paired synthetic underwater data. Third, we adapt to real imagery using rewatering consistency and scale-invariant relative-depth supervision while retaining synthetic replay. The DINOv3 tokenizer and pose teacher remain frozen throughout; the Gaussian and Water heads are optimized at the base learning rate, whereas the remaining backbone parameters use a reduced rate. Detailed schedules and loss weights are provided in the supplementary material.

4 Experiments

4.1 Experimental Setup

Data and splits. Training combines paired synthetic data from TartanAir (Wang et al. 2020) and Hypersim (Roberts et al. 2021), degraded with Eq. 2, with real underwater data from FLSea (Randall and Treibitz 2023) and LabPool, which we recorded at 30 Hz in a controlled laboratory pool. Synthetic data provide clean images, depth, poses, and medium parameters; FLSea provides underwater imagery, photogrammetric depth, and camera poses. FLSea comprises eight scenes across distinct water environments; we train and adapt on six scenes, while Cross Pyramid Loop and Flatiron are completely held out for evaluation, with no frames from either test scene used during training or adaptation.

Protocol and baselines. We use one frozen checkpoint for all principal results and perform no test-time fine-tuning or scene-specific optimization. Novel-view synthesis is evaluated for $N \in \{3, 6, 16, 32, 48, 64\}$ unposed context views; the main paper reports representative budgets and the full sweep is in the supplement. Target images are excluded from reconstruction, and target-camera alignment is estimated from context cameras only. We compare per-scene 3DGS baselines, which require COLMAP and optimization, with the feed-forward AnySplat (Jiang et al. 2025), and NoPoSplat (Ye et al. 2025).

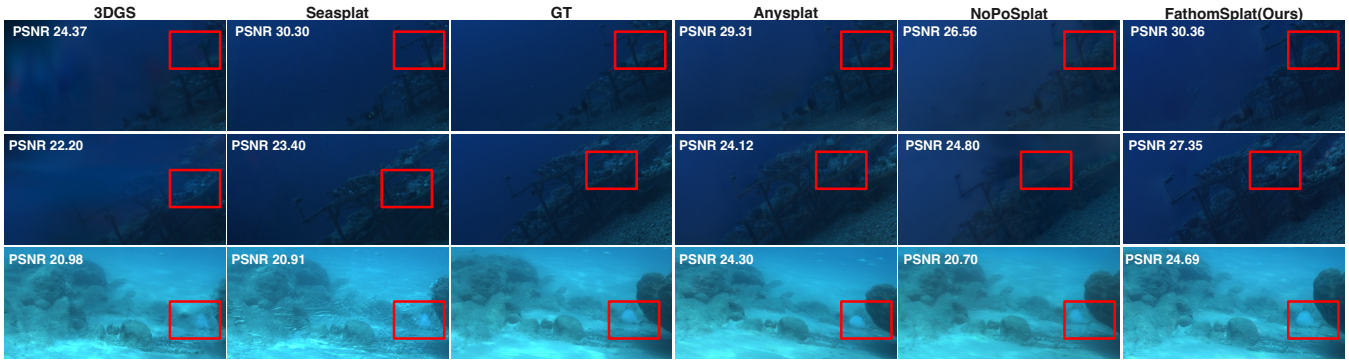


Figure 3: Qualitative novel-view synthesis on held-out underwater scenes. All methods render the same target views from the designated context images.

	N=3				N=16				N=48				N=64			
Method	P	S	L	T	P	S	L	T	P	S	L	T	P	S	L	T
Per-scene optimization																
3DGS	19.69	0.579	0.361	102.0	21.80	0.608	0.312	112.3	17.55	0.455	0.468	147.6	19.28	0.516	0.400	200.1
SeaSplat	20.46	0.516	0.390	622.6	21.34	0.530	0.359	575.0	19.12	**0.551**	0.427	570.4	19.94	**0.572**	**0.378**	623.7
Feed-forward																
AnySplat	17.82	0.433	0.488	**0.13**	19.22	0.466	0.437	0.23	18.81	0.502	0.467	0.68	19.01	0.500	0.466	0.99
NoPoSplat	**20.95**	0.510	0.416	0.19	20.99	0.506	0.409	0.24	**20.74**	0.536	0.444	**0.25**	20.17	0.520	0.465	**0.27**
FathomSplat (*Ours*)	20.66	**0.622**	**0.359**	0.17	**21.92**	**0.616**	**0.297**	**0.20**	20.56	0.537	**0.420**	0.66	**20.41**	0.530	0.441	0.94

Table 1: Novel-view synthesis averaged over the held-out FLSea sequences Cross Pyramid Loop and Flatiron at four input budgets. P, S, L, and T denote PSNR↑, SSIM↑, LPIPS↓, and time in seconds↓. Best results are bold and second best results are underlined. The complete six-budget sweep is reported in the supplement.

Metrics and implementation. We report PSNR, SSIM, LPIPS, and end-to-end reconstruction time for novel views; pose AUC; median-aligned depth AbsRel and δ_1; and clean-render and medium-estimation metrics for the decomposition study. FathomSplat uses a VGGT-Ω (Wang et al. 2026) backbone, an AnySplat-style Gaussian head, and a Water-Head. Training schedules, data mixtures, alignment, timing, and baseline details are provided in the supplement.

4.2 Comparison with State of the Art

Table 1 averages results over the held-out Cross Pyramid Loop and Flatiron sequences and reports N=3, 16, 48, 64; the complete six-budget sweep is provided in the supplement. FathomSplat gives the best SSIM and LPIPS at N=3, the best quality across all three metrics at N=16, and the best LPIPS at N=48, where it also ranks second in PSNR and SSIM. At N=64, it gives the best PSNR. Reconstruction takes 0.17–0.94 s across the reported budgets. Figures 3 and 4 show the corresponding 2D and 3D results.

4.3 Camera Pose Estimation

We evaluate relative camera pose accuracy on the held-out FLSea sequences Cross Pyramid Loop and Flatiron, using 10 frames sampled with a fixed seed inside a local temporal window. Following (Jiang et al. 2025), Table 2 reports AUC of the relative pose error at 5, 10, 20, and 30 degrees.

	CPL				FI			
Method	@30	@20	@10	@5	@30	@20	@10	@5
AnySplat	0.888	0.832	0.664	0.360	0.259	0.008	0.000	0.000
NoPoSplat	0.326	0.157	0.018	0.000	0.056	0.006	0.000	0.000
FathomSplat	**0.923**	**0.884**	**0.769**	**0.556**	**0.831**	**0.747**	**0.496**	**0.124**

Table 2: Camera pose AUC↑ on held-out FLSea sequences using 10 frames. CPL and FI respectively denote Cross Pyramid Loop and Flatiron. AnySplat and FathomSplat predict all cameras jointly; NoPoSplat uses pairwise inference followed by PnP.

4.4 Depth Estimation

Table 3 compares dense context-view depth against feed-forward baselines, with a qualitative example shown in Figure 5. The synthetic column averages 12 held-out Hypersim and 8 held-out TartanAir two-view windows; the FLSea column averages 8 windows from the held-out Cross Pyramid Loop scene. Every method receives the same first/last context pair, exported at 448×448, and is scored with the same ground-truth validity mask. We estimate one median scale jointly over the two views of each window, compute AbsRel and δ_1, and then average per-window scores. FathomSplat therefore predicts geometrically consistent scene structure

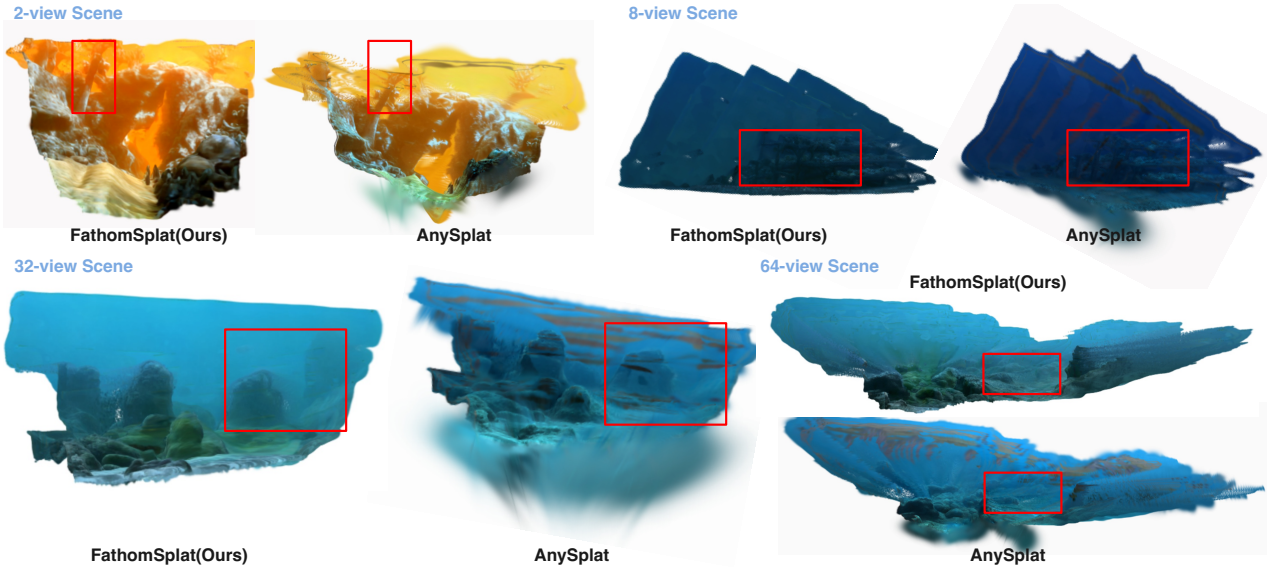


Figure 4: Qualitative 3D Gaussian reconstruction from sparse unposed underwater views. The reconstructed scenes are visualized from matched viewpoints.

up to one global scale rather than metric depth.

Method	Synthetic		FLSea	
	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑
AnySplat	0.105	0.875	0.102	0.882
NoPoSplat	0.347	0.580	0.185	0.702
FLARE	0.518	0.433	0.287	0.468
FathomSplat (*Ours*)	**0.079**	**0.937**	**0.046**	**0.990**

Table 3: Relative-depth accuracy on held-out synthetic and FLSea data. AbsRel↓ and δ_1 ↑ use one median alignment per two-view window.

FathomSplat ranks first on both domains and both metrics. Relative to AnySplat, it reduces AbsRel by approximately 25% on synthetic data and 55% on held-out FLSea, while improving δ_1 by 6.2 and 10.8 percentage points. On the separately evaluated held-out Flatiron sequence, it obtains an AbsRel/δ_1 of 0.043/0.983.

4.5 Physics Decomposition and Ablations

Table 4 isolates the medium–scene decomposition under a matched feed-forward protocol on held-out synthetic scenes. Real data lack paired clear references and measured medium parameters, so medium-free recovery is evaluated quantitatively only on synthetic data. The evaluation uses 12 Hypersim and 8 TartanAir windows, with identical samples and sampled medium parameters across all variants. To evaluate the geometry prior, we replace both the VGGT-Ω backbone and frozen teacher with the original VGGT while keeping all other training settings unchanged. This change reduces clean and rewatered PSNR by 1.50 and 1.14 dB, respectively, while medium MAE and AbsRel remain nearly unchanged. Ablating individual objectives further confirms the role of each term. Dropping $\mathcal{L}_{medium}$ increases the medium-parameter error from 0.13 to 0.20, dropping $\mathcal{L}_{rewater}$ reduces re-watered PSNR from 24.97 to 24.00 dB, and dropping $\mathcal{L}_{clear}$ lowers clean PSNR. Additional training-strategy ablations, including the backbone-update study, are reported in the supplementary material.

Variant	Clean		ReWater		MAE↓	AbsRel↓
	P↑	L↓	P↑	L↓		
VGGT	21.18	0.459	23.83	0.424	0.132	0.078
w/o $\mathcal{L}_{clear}$	21.93	0.302	24.87	**0.252**	0.132	0.079
w/o $\mathcal{L}_{rewater}$	22.10	0.367	24.00	0.347	0.132	0.079
w/o $\mathcal{L}_{medium}$	21.98	0.312	24.44	0.258	0.203	**0.077**
Full (Ours)	**22.68**	**0.298**	**24.97**	0.256	**0.132**	**0.077**

Table 4: Architecture and physics decomposition ablations on held-out synthetic scenes. P and L denote PSNR and LPIPS. Clean and ReWater evaluate medium-free and observation-domain rendering; MAE evaluates water parameters, and AbsRel is median-scaled.

Table 5 compares matched variants on held-out FLSea, averaging eight windows from each sequence. Full uses re-watering, whereas No Water directly renders the observation domain. Full improves PSNR at every budget and is best on all three metrics with three and 64 input views; with 16 input views, it retains a 1.19 dB PSNR gain but trades off SSIM and LPIPS.

4.6 Real-World Adaptation

Effect of the training-data mix. Table 6 isolates the contribution of each training-data source on held-out FLSea Cross Pyramid Loop across input-view budgets, using a single frozen architecture and feed-forward inference. Starting

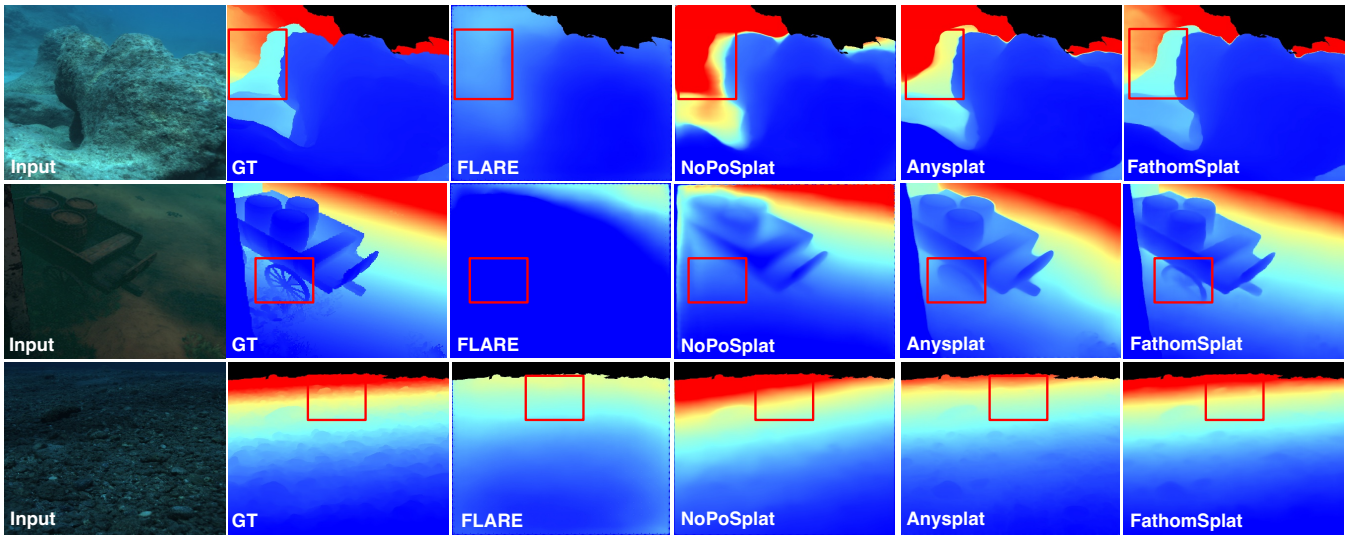


Figure 5: Qualitative relative-depth comparison on a held-out underwater scene. All methods use the same input views; quantitative results are reported in Table 3.

	N=3			N=16			N=64		
Variant	P	S	L	P	S	L	P	S	L
No Water	20.46	0.506	0.397	20.98	0.530	0.349	20.56	0.529	0.429
Full	**20.64**	**0.516**	**0.390**	**22.17**	**0.505**	**0.374**	**20.68**	**0.537**	**0.420**

Table 5: Water-Head ablation on held-out FLSea. P↑, S↑, and L↓ denote PSNR, SSIM, and LPIPS.

from the in-air geometry prior, *synthetic* data alone offer no consistent real-scene gain. Adding *real* underwater data gives the best PSNR and SSIM at every budget and the best LPIPS at three and six views; the in-air prior remains lower in LPIPS at 64 views. Real-domain adaptation therefore improves fidelity consistently across input-view budgets, with a perceptual trade-off at the largest budget.

Data		N=3			N=6			N=64		
Syn.	Real	P	S	L	P	S	L	P	S	L
		20.30	0.523	0.381	19.86	0.522	0.364	22.42	0.531	**0.372**
✓		19.76	0.534	0.440	20.74	0.534	0.430	20.92	0.531	0.503
✓	✓	**21.70**	**0.541**	**0.372**	**22.62**	**0.543**	**0.360**	**22.63**	**0.531**	0.401

Table 6: Training-data ablation on Cross Pyramid Loop, a held-out FLSea sequence. Syn. and Real indicate synthetic and real underwater adaptation data. P↑, S↑, and L↓ denote PSNR, SSIM, and LPIPS.

5 Conclusion

We presented FathomSplat, a physics-guided feed-forward framework for reconstructing medium-free 3D Gaussian scenes from sparse, unposed underwater images. The model jointly predicts camera parameters, scene geometry and appearance, and per-view medium parameters, then couples these predictions through a differentiable rewatering operator. This explicit factorization assigns attenuation and backscatter to the observation model rather than absorbing them into Gaussian appearance and geometry. Paired synthetic supervision constrains the decomposition, while rewatering consistency, relative-geometry supervision, and synthetic replay support adaptation to real underwater imagery. Experiments show that FathomSplat achieves the best performance across all three novel-view metrics at the representative 16-view setting, the strongest relative-depth and camera-pose results among the evaluated feed-forward methods, and reconstruction times between 0.17 and 0.94 seconds. Ablations further support the role of the scene and medium factorization and its training objectives. The current formulation uses a compact per-view medium model and does not fully capture forward scattering, nonuniform illumination, or dynamic particles. Extending the model to spatially varying and dynamic water conditions is an important direction for future work.

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